# PIRACETAM

Limited evidence
Renal dose adjustment

**CHEMICAL NAME** 2-(2-oxopyrrolidin-1-yl)acetamide
**CLASS** Nootropic (cyclic GABA derivative)
**TRADE NAMES** Nootropil · Lucetam · Memotropil · Piracetam (generic)

## THERAPEUTIC USES / INDICATIONS

- Cortical myoclonus (adjunct) — the best-supported indication.
- Cognitive impairment/vertigo, post-stroke aphasia, dyslexia (marketed in some countries; evidence weak).

## PHARMACODYNAMICS

- Modulates **neuronal membrane fluidity** and neurotransmission (cholinergic, glutamatergic); ↓ platelet aggregation and red-cell adhesion; improves microcirculation.
- Exact mechanism uncertain.

## ADMINISTRATION

- **PO** tablets or solution in 2–3 divided doses, with or without food.
- **IV** in some countries.
- Taper gradually in myoclonus (sudden withdrawal → seizures).

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Nervousness, hyperkinesia, insomnia.
- Weight gain.
- Depression, somnolence, asthenia.

### ADVERSE EFFECTS (SERIOUS)

- Bleeding (antiplatelet effect); caution with anticoagulants.
- Seizures on abrupt withdrawal.
- Hypersensitivity (rare).

## DOSAGES

- **Cortical myoclonus:** 7.2 g/day in 2–3 doses, increased by 4.8 g/day every 3–4 days; max 24 g/day.
- **Cognitive disorders (where licensed):** 2.4–4.8 g/day.
- **CrCl 50–79:** ⅔ dose; 30–49: ⅓ dose; 20–29: ⅙ dose; contraindicated if CrCl <20.

## CONTRAINDICATIONS & CAUTIONS

- End-stage renal disease (CrCl <20).
- Cerebral haemorrhage.
- Huntington's chorea.
- **Caution:** bleeding disorders, major surgery.

### KEY INTERACTIONS

- **Warfarin, antiplatelets:** ↑ bleeding risk.
- **Thyroid hormones:** confusion, irritability reported.

## PHARMACOKINETICS

- **Peak:** 30–60 min. Bioavailability ~100%.
- Not metabolised.
- Excreted unchanged in urine. **t½:** 5 h.

## NURSING MANAGEMENT

- Check renal function for dose.
- Monitor for nervousness, insomnia, bleeding.
- Assess myoclonus frequency.
- Taper slowly when stopping.

## PATIENT EDUCATION

- Take in divided doses.
- Report unusual bleeding or bruising.
- Do not stop suddenly if used for myoclonus.
- Benefit for memory is uncertain.

**KEY POINTS**

- Nootropic; best evidence for cortical myoclonus.
- Nervousness, insomnia, weight gain.
- Renally excreted — adjust dose.
- Antiplatelet effect.